

The complete practical program: 12 labs covering terminology, workflow, gene flow, non-target analysis, resistance simulation, exposure, composition, dose-response, risk matrices, uncertainty, regulatory role-play and the capstone

Simulation, computational and case-study formats - no wet-lab GMO work;
instructor keys included

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All datasets used in this package are simulated teaching data, clearly labelled as such. This course involves no wet-lab work with living GMOs; case studies and regulatory summaries reflect published evidence and should be verified against current primary sources.

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Lab 01 — GMO Risk-Assessment Terminology and Hazard Identification

Activity type: Case Study / Simulation (no wet lab)

Duration: 2 hours · **Level:** Beginner

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■ [Related Module: Module 2 — Hazard, Risk and Exposure](#) · ■ [Assessment: MCQs](#)

Learning objectives

By the end of this practical you will be able to:

1. Define hazard, exposure, risk, uncertainty, risk management, and risk communication without notes.
2. Distinguish hazards from risks in statements commonly heard in public debate.
3. Identify potential hazards for a described GMO using a structured worksheet.
4. Express identified hazards as testable statements (mechanism, exposure pathway, consequence).

Background

Public discussion of GMOs constantly confuses "hazard" (a thing that could cause harm) with "risk" (the chance × consequence of that harm under real conditions). Professional risk assessment begins by separating them — and by phrasing every identified hazard so that evidence could, in principle, support or refute it.

Scientific principle

The conceptual chain taught in Module 2:

Hazard + Exposure + Dose/Response + Context → Risk (conceptually)

A hazard statement has three parts: **agent** (what could cause harm), **mechanism** (how), and **endpoint** (what harm). If any part is missing, the statement is not yet assessable.

Materials/data

- Printed/quoted public statements (provided in the worksheet below).
- The hazard-identification worksheet (Section 8).
- Course Cheat Sheet for terminology reference.
- No laboratory equipment, living material, or biohazard is involved.

Safety

This is a **paper exercise**: no biosafety precautions are required. The relevant professional safety here is intellectual: never treat an advocacy statement as evidence.

Step-by-step workflow

1. **Terminology drill (15 min).** Without notes, write your own one-sentence definitions of: hazard, exposure, risk, uncertainty, risk management, risk communication. Compare with the Cheat Sheet; correct your own.
2. **Statement sorting (25 min).** Classify each statement below as: (H) hazard claim, (R) risk claim, (M) management claim, (C) communication/perception claim.

3. **Hazard re-writing (25 min).** For each (H) statement, rewrite it as a complete hazard statement (agent–mechanism–endpoint).
4. **Worksheet (30 min).** Complete the hazard-identification worksheet for the crop described in Section 7.
5. **Peer review (15 min).** Swap worksheets; mark each hazard statement assessable or not, with reasons.
6. **Debrief (10 min).** Instructor aggregates hazards; class maps them to course modules.

Data tables

Statement set for Step 2:

#	Statement	Type (H/R/M/C)?
1	"Bt toxin is an insecticidal protein — it is a biological pesticide."	
2	"Bt maize pollen reaching milkweed at field-edge densities is unlikely to reach doses harming monarch larvae."	
3	"Farmers should plant 20% non-Bt refuges to slow resistance."	
4	"People are worried because they cannot see the genes that were changed."	
5	"Herbicide-tolerant crops could interbreed with wild relatives."	
6	"Herbicide-tolerant volunteers are controlled by an alternative herbicide in the rotation."	
7	"GM fish escaping into rivers could outcompete wild salmon."	
8	"Surveys show consumers overestimate pesticide residues on GM crops."	

Calculations

None in this lab. (Quantitative treatments begin in Lab 03.)

Expected results

- Statements 1, 5, 7 are hazard claims (though statement 1 lacks a mechanism/endpoint — the rewrite step exposes this).
- Statements 2, 6 are risk/management hybrid claims depending on reading; defensible either way **if** justified.
- Statements 3, 6 contain management actions.
- Statements 4, 8 are communication/perception claims — legitimate topics (Module 21), but not science claims.

Interpretation

A hazard claim without exposure or context is not a risk claim — and a perception claim is not evidence about either. Most public "GMO debates" are people arguing about different boxes of this taxonomy at cross purposes.

Troubleshooting

Not applicable (paper exercise). If groups disagree on a classification, the productive question is: *what evidence would move this statement from one box to another?*

Questions

1. Rewrite statement 1 as a fully assessable hazard statement.
2. Why is statement 2 a risk claim while statement 1 is a hazard claim?
3. Give one example where a communication claim influenced real regulatory outcomes in this course's case studies.
4. Which of your worksheet hazards maps to Module 6, and how would you make it testable?

Viva questions

1. Define "hazard" and "risk" in one sentence each.
2. Why does "absence of evidence is not evidence of absence" matter for hazard identification?
3. Name the three parts of a complete hazard statement.
4. Why is "GM crops are dangerous" not an assessable statement?

Instructor answer key

- Q1 example: "The Cry1Ac protein (agent), when ingested by non-target lepidopteran larvae feeding on pollen-dusted plants near Bt cotton fields (mechanism/exposure), may reduce their survival (endpoint)."
- Q2: Statement 2 contains an exposure estimate (field-edge pollen density) and a dose comparison — that is risk logic; statement 1 identifies a property only.
- Q3: StarLink (Case 5) — feed-only approval collapsed under public/consumer exposure fears; monarch episode (Case 2) — perception preceded exposure data.
- Q4: "Dangerous" names no agent, mechanism, or endpoint; nothing about it can be supported or refuted by data as phrased.

Advanced challenge

Take the class's best hazard from the worksheet and design the *minimum* observation or experiment that would falsify it. State the falsification criterion numerically.

Hazard-identification worksheet (Step 4)

GMO scenario: A cotton variety engineered with **two** Bt toxins (Cry1Ac + Cry2Ab) for bollworm control, to be cultivated in a region with irrigated cotton, some feral cotton plants near gins, and three years of prior single-toxin Bt cotton use.

#	Potential hazard (agent–mechanism–endpoint)	Exposure pathway	Module	Priority (H/M/L) + why
1				
2				
3				
4				
5				

Expected worksheet entries (instructor): resistance to Cry1Ac (used three years — Module 11); cross-resistance/stack failure if resistance alleles present (Module 25.7); effects on non-target predators via intoxicated larvae (Module 8); gene flow to feral cotton near gins (Module 6); soil-protein exposure from residue (Module 9). Students may legitimately differ on priority ranking if reasons are given.

Previous: (start of labs)* · *Next:* Lab 02 — Risk-Assessment Workflow*

Lab 02 — Risk-Assessment Workflow Exercise

Activity type: Simulation (framework walk-through on a realistic dossier excerpt)

Duration: 2.5 hours · **Level:** Beginner → Intermediate

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Next Lab](#)

■ **Related Module:** [Module 3 — The GMO ERA Framework](#) · ■ **Data:** [Risk-matrix scores](#)

Learning objectives

1. Execute every stage of the ERA framework on one concrete GMO.
2. State the inputs, questions, and outputs of each framework stage.
3. Produce a defensible problem-formulation document.
4. Recognise where assessments fail (missing stage, missing comparator, unexamined uncertainty).

Background

Module 3 presents the ERA spine: problem formulation → hazard identification → hazard characterization → exposure assessment → risk characterization → uncertainty analysis → risk management → monitoring → communication. Reading it is easy; executing it on a real organism with incomplete information is the professional skill. This lab forces the execution with a deliberately incomplete dossier.

Scientific principle

Each stage has defined inputs and outputs; a workflow is only as good as its weakest hand-off. Problem formulation converts policy protection goals into **assessment endpoints** and **measurement endpoints** — the step most skipped by beginners and most demanded by regulators.

Materials/data

- Dossier excerpt (Section 6) — fictional but realistic: *InsectGuard*[™] maize, expressing a novel insecticidal protein "IP-47" against stem borers.
- Course Cheat Sheet (framework diagram).
- Module 3 chapter as reference.
- Worksheet in Section 8.

Safety

Paper exercise; no biosafety requirements.

Step-by-step workflow

1. **Read the dossier (15 min).** Note what is present and absent.
2. **Problem formulation (30 min).** Complete worksheet Part A: protection goals → assessment endpoints → measurement endpoints → hypotheses.
3. **Hazard identification (20 min).** Part B: list candidate hazards with agent–mechanism–endpoint.
4. **Hazard & exposure characterization (30 min).** Part C: for your top two hazards, specify what data would quantify them, and what exposure pathways apply in the described receiving environment.
5. **Risk characterization sketch (20 min).** Part D: qualitative risk sentence per hazard using the likelihood × consequence × uncertainty language of Module 17.
6. **Gap analysis (15 min).** Part E: list every stage where the dossier's data are insufficient; write the one study you would commission first, with its design.

7. **Debrief (10 min).** Compare gap lists; identify the most commonly missing stage (instructor: it is usually uncertainty analysis).

Data tables

Dossier excerpt — InsectGuard™ maize (fictional):

- Event: single-copy insertion, intact ORF confirmed; expression highest in pollen and pith (µg/g values "pending").
- Protein IP-47: novel insecticidal protein; sequence shows no homology to known allergens/toxins beyond a shared motifs database hit of unclear function; digestibility assay "in progress".
- Agronomy: 2-year yield trials at 6 sites; no non-target studies commissioned; no compositional analysis.
- Receiving environment: irrigated maize valleys; wild relative *Tripsacum*-like grasses present at field margins; beekeeping active in valley; stem-borer pressure high.
- Management proposal: none yet.

Worksheet:

Part	Item	Your entry
A1	Protection goal(s) implied by the receiving environment	
A2	Assessment endpoint(s)	
A3	Measurement endpoint(s) — what you would actually measure	
A4	Risk hypothesis(es)	
B1–B4	Candidate hazards (agent–mechanism–endpoint)	
C1–C2	Data needed to quantify top hazards; exposure pathways	
D1–D2	Qualitative risk statements	
E1	Dossier gaps by framework stage	
E2	First study you commission (design in 3 lines)	

Calculations

None — this lab builds the structure that later labs quantify.

Expected results (instructor)

- Part A: protection goals should include non-target biodiversity (pollinators, natural enemies), soil function, wild-relative genetic integrity; measurement endpoints might be adult bee survival/colony strength, predator abundance in plots, outcrossing frequency at margins.
- Part B: strong entries include IP-47 toxicity to non-target insects with shared target-receptor biology; gene flow + establishment in wild grasses; allergenicity/IP-47 digestibility unknown.
- Part C: exposure = pollen dose to larvae/bees; expression values are the missing datum ("pending" — students must flag it).
- Part D: correct statements name likelihood, consequence, and uncertainty explicitly and conditionally ("low–moderate pending expression data").
- Part E: expected gaps — no expression quantification, no non-target tier-1 tests, no compositional analysis, no monitoring plan, no uncertainty analysis, no management proposal.

Interpretation

The dossier is not unusual — real submissions have exactly these gaps at draft stage. The professional act is not rejecting it but specifying, via problem formulation, which data would change the risk estimate.

Troubleshooting

- *Groups produce vague endpoints ("environment is safe")*. Push: an endpoint must be a measurable entity + attribute (e.g., "abundance of the predator *Orius* spp. per plant").
- *Hazards duplicate the intended effect*. The intended pest effect is a benefit, not a hazard — the hazard is the same protein reaching *non*-targets.

Questions

1. Why can expression "values pending" block the entire exposure assessment?
2. Distinguish an assessment endpoint from a measurement endpoint with your own example.
3. Which dossier absence is most likely to invalidate a food-safety conclusion?
4. Why is "no management proposal" itself a finding?

Viva questions

1. What does problem formulation convert, into what?
2. Name the nine framework stages in order.
3. Why is uncertainty analysis a stage, not an afterthought?
4. What makes a risk hypothesis testable?

Instructor answer key

- Q1: Exposure = dose × pathway; without expression values no dose can be estimated for any non-target or consumer.
- Q2: Assessment endpoint = the valued ecological property (e.g., "survival of native butterfly larvae"); measurement endpoint = the proxy you measure (e.g., "larval survival on pollen-dusted host leaves at 25 m").
- Q3: Compositional analysis + protein digestibility — the classic novel-food pair.
- Q4: Without management conditions the assessment cannot attach conditions (refuges, buffers) that make residual risk acceptable; "acceptable risk under conditions" is the normal regulatory outcome.
- Viva: (1) policy protection goals → operational assessment endpoints/hypotheses. (2) Problem formulation, hazard ID, hazard characterization, exposure assessment, risk characterization, uncertainty analysis, risk management, monitoring, communication. (3) It determines what confidence the conclusion can carry and what monitoring must watch. (4) It must be falsifiable by named data.

Advanced challenge

Redesign the dossier's field-trial programme so that, after two seasons, the exposure assessment for bees is closure-ready. Specify sites, plots, and the single measurement that matters most.

Previous: [Lab 01](Lab-01-Terminology-and-Hazard-Identification.md) · Next: [Lab 03](Lab-03-Gene-Flow-Data-Analysis.md)

Lab 03 — Gene-Flow Data Analysis

Activity type: Computational Exercise

Duration: 2.5 hours · **Level:** Intermediate

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Next Lab](#)

■ **Related Module:** [Module 6 — Gene Flow and Introgression](#) · ■ **Dataset:** [Exercise 1 — Gene flow](#) · ■ **Assessment:** [Data-Interpretation Questions](#)

Learning objectives

By the end of this practical you will be able to:

1. Summarise hybrid-frequency data by distance and quantify uncertainty.
2. Fit and interpret a negative-exponential decay model of pollen-mediated gene flow.
3. Translate a fitted decay curve into a risk-management statement (isolation distance vs threshold).
4. Critique the dataset's limitations for regulatory extrapolation.

Background

Pollen-mediated gene flow from a transgenic crop to non-GM fields or wild relatives typically decays with distance. Risk assessors use this decay to set isolation distances and buffer zones. The assessment question is never "is gene flow zero?" (it never is) but "below what frequency does consequence become negligible, and at what distance does the curve cross that threshold?"

Scientific principle

Empirically, outcrossing frequency $f(d)$ versus distance d is well approximated by a decaying exponential (plus a long tail that field data barely capture):

$$f(d) = a \cdot \exp(-b \cdot d)$$

Parameters: a = near-field (edge) frequency; b = decay constant. **Risk-management logic:** the buffer zone works on the steep part of the curve; the flat tail means zero flow is unattainable, so "acceptable frequency" is a management threshold — a judgment, informed by the biology.

Materials/data

- DATA/gene-flow/gene_flow_by_distance.csv — simulated hybrid frequencies at 1, 5, 10, 25, 50, 100 m from the pollen source; 12 replicate plots per distance. **Clearly labelled simulated data.**
- Python 3 with numpy/pandas/matplotlib.
- Reference implementation: DATA/analysis_demo.py (Exercise 1 section).

Safety

No biosafety concerns (computational exercise on simulated data).

Step-by-step workflow

1. **Load and inspect** — check columns, units, missing values, n per group.
2. **Summarise** — mean hybrid frequency and standard error per distance (table below).
3. **Visualise** — scatter of raw values with mean $\pm$ 95% CI per distance, log-y axis.
4. **Fit** — log-linearise and fit $\ln f = \ln a - b \cdot d$; report R^2 .
5. **Threshold analysis** — find the distance where the fitted curve's upper 95% bound falls below 0.001 (0.1%).

6. **Interpret** — write the management sentence the curve supports.
7. **Challenge** — repeat with a half-size source field; how do a and b move?

Data tables

Table 1 (complete during Step 2):

Distance (m)	Mean frequency	SE	n
1			
5			
10			
25			
50			
100			

Table 2 (Step 5):

Threshold	Distance where upper 95% bound < threshold
0.001	

Calculations

- Wilson or normal-approximation 95% CI per distance (state which and why).
- Least-squares fit on log-transformed data: $\ln(a)$, b , R^2 .
- Inverse prediction: $d(0.001) = \ln(a/0.001)/b$.

Expected results

(Simulated dataset — documented by `analysis_demo.py`.) Frequency falls from ≈ 0.030 at 1 m to ≈ 0.0002 at 100 m; the fit is strongly exponential ($R^2 > 0.95$ on log scale); the 0.1% threshold is crossed between roughly 15–30 m for the *point* estimate, later for the upper bound.

Interpretation

- A buffer of ~ 30 m makes *typical* gene flow below 0.1%; making it below 0.1% with **95% confidence** requires more distance (or wider buffers) — report both, not just the point estimate.
- The tail matters: seed spillage and machinery transfer bypass pollen distance entirely, so isolation distance alone is never sufficient (Module 6).

Troubleshooting

Symptom	Cause	Fix
Fit curve bends on log plot	Non-exponential tail; plot suggests power-law	Report both models; use empirical interpolation for thresholds
Negative predicted frequencies	Fitting raw (not log) data	Fit on log scale; back-transform
CI crossing zero	Small n at far distances	Use Wilson interval; report n explicitly

Questions

1. Why fit on the log scale for this dataset?

2. At 100 m the minimum observed frequency is 0. Is that evidence of *absence* of gene flow? Explain using "not detected" vs "absent."
3. Your upper-bound distance is $\sim 1.6\times$ the point-estimate distance. What does that gap tell the risk manager?
4. Which unmeasured variable most limits extrapolation to a real landscape, and why?

Viva questions

1. Define pollen-mediated gene flow.
2. Why can buffer zones never reduce gene flow to zero?
3. What is introgression, and how does it differ from first-generation hybridisation?
4. Name two non-pollen routes of transgene movement.

Instructor answer key

- Q1: Variance scales with the mean and the relationship is multiplicative; log-linearising makes the exponential fit a simple linear regression with well-behaved errors.
- Q2: No — zero observations at that distance is consistent with frequencies below the detection limit given $n=12$; absence of detection $\neq$ absence of flow (distinguish statistically).
- Q3: The manager's confidence, not just the expected value, determines compliance risk; a rule set at the point estimate fails one field in twenty by construction.
- Q4: Flowering synchrony (temporal overlap) — it can dominate frequency and varies year to year; the dataset holds it constant.

Advanced challenge

Simulate a landscape of 50 recipient fields at random distances using your fitted (a, b), add Poisson sampling noise, and estimate the probability that **at least one** field exceeds the 0.1% threshold. How does that landscape-scale probability change the management conclusion compared with single-field analysis?

Previous: [Lab 02 — Risk-Assessment Workflow](Lab-02-Risk-Assessment-Workflow.md) · Next: [Lab 04 — Non-Target Data Analysis](Lab-04-Non-Target-Data-Analysis.md)

Lab 04 — Non-Target Organism Data Analysis

Activity type: Computational Exercise

Duration: 2.5 hours · **Level:** Intermediate

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Next Lab](#)

■ [Related Module: Module 8 — Non-Target Organisms](#) · ■ [Dataset: Exercise 2 — Non-target survival](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Estimate survival proportions with defensible confidence intervals (Wilson).
2. Distinguish *statistical* from *biological* significance using effect size + interval + comparator.
3. Explain why a laboratory hazard finding becomes a risk claim only after exposure quantification.
4. Compare an effect against the correct ecological comparator (insecticide regime, not idealised unsprayed crop).

Background

Non-target assessment (Module 8) proceeds in tiers: laboratory toxicity → semi-field → field. A common failure mode is stopping at the laboratory stage, where doses are high and exposure is forced. The analysis taught here — effect size, interval, comparator — is the analytical core of every tier.

Scientific principle

For survival data, each treatment group gives an estimate $p = \text{alive}/n$. The **Wilson score interval** behaves well at small n and extreme proportions:

$$CI = [p + z^2/2n \pm z \cdot \sqrt{ p(1-p)/n + z^2/4n^2 }] / (1 + z^2/n)$$

The ERA question is then threefold: (1) is the interval compatible with no effect? (2) is the *point estimate* biologically meaningful at field exposure? (3) compared with what alternative — the insecticide regime the Bt crop replaces?

Materials/data

- DATA/non-target/ — simulated survival of a beneficial predator and a parasitoid under: control diet, Bt-protein diet (high dose), Bt-pollen diet (field-realistic), conventional-insecticide exposure (positive realism control). $n=200$ per group. **Simulated data.**
- Python 3 (numpy, pandas, matplotlib); reference: DATA/analysis_demo.py Exercise 2.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Load** survival counts; build the treatment × species table.
2. **Estimate** survival per group with Wilson 95% CIs.
3. **Contrast** Bt-pollen (field-realistic) vs control: difference in proportions + interval.
4. **Comparator analysis:** same contrast vs the conventional-insecticide group.
5. **Exposure mapping:** assign each treatment to its ERA tier (forced-dose lab / realistic pollen / field regime) and re-plot as a tiered figure.
6. **Write** the three-sentence risk statement (estimate → uncertainty → comparator).

7. **Challenge:** power analysis — what n would detect a true 3-point survival drop with 80% power?

Data tables

Species	Treatment	n	Alive	Survival	Wilson 95% CI
Predator	Control				
Predator	Bt high-dose				
Predator	Bt pollen (field)				
Predator	Conventional insecticide				
Parasitoid	(same four rows)				

Calculations

- Wilson CI per group (show one hand calculation, then automate).
- Difference in proportions with Newcombe-style interval (or bootstrap) for Bt-pollen vs control.
- Power calculation for the 80%/3-point detection (normal approximation acceptable; state it).

Expected results

(Simulated — per `analysis_demo.py`.) Bt high-dose may show a small, interval-compatible reduction; Bt pollen (field-realistic) indistinguishable from control; conventional insecticide shows a large, unambiguous reduction. The documented teaching pattern: **hazard at forced dose, no risk at realistic exposure, comparator regime is worse.**

Interpretation

The three-sentence template students must produce, e.g.:

"Predator survival under field-realistic Bt pollen was 0.93 vs 0.93 in controls (difference -0.003 , 95% CI -0.06 to $+0.05$), compatible with no effect. The high-dose laboratory treatment suggests a hazard exists at sufficient dose. Under the comparator regime (single conventional insecticide application), survival fell to 0.62 — so the relevant ecological comparison favours the Bt system for this endpoint."

Note what this does *not* say: it does not say "Bt is safe for all non-targets" — one species, one route, one season.

Troubleshooting

Symptom	Cause	Fix
CI very wide	n too small per group	Report n ; avoid over-interpretation; power analysis
"Significant" tiny difference	Very large n	Report effect size first; ask biological relevance (Module 16)
Comparator treated as 'control'	Design confusion	The comparator is a <i>treatment</i> , not a control; label clearly

Questions

1. Why is the conventional-insecticide group essential to the risk conclusion?
2. The high-dose group shows a small drop with CI $[-0.08, -0.01]$. Does this establish field risk? Why not?
3. What would make the parasitoid's exposure pathway different from the predator's?
4. Why is "no significant difference" not the same as "no effect"?

Viva questions

1. Define non-target organism and give two examples with different trophic positions.
2. What is a tritrophic exposure pathway?
3. Why do regulatory non-target studies use maximum hazard dosing at tier 1?
4. Name two field-level confounders in non-target monitoring.

Instructor answer key

- Q1: It supplies the ecological counterfactual — the regime Bt replaces; without it "small or no Bt effect" has no decision meaning.
- Q2: No — forced-dose exposure has no field counterpart; risk requires exposure quantification (Module 2/8); tier 1 is a hazard screen.
- Q3: Parasitoids develop inside hosts — exposure via host tissue/haemolymph (a tritrophic route), whereas the predator consumes prey whole; timing and dose differ.
- Q4: Underpowered test; CI may include large effects; equivalence needs an equivalence margin, not a failed significance test.

Advanced challenge

Using the same simulated machinery, generate 1,000 bootstrap datasets and report the probability that the Bt-pollen effect appears "significant" ($p < 0.05$) in at least one of ten endpoints — the garden of forking paths quantified. Connect to multiplicity in real dossiers.

Previous: [Lab 03 — Gene-Flow Data Analysis](Lab-03-Gene-Flow-Data-Analysis.md) · Next: [Lab 05 — Resistance Evolution Simulation](Lab-05-Resistance-Evolution-Simulation.md)

Lab 05 — Resistance Evolution Simulation

Activity type: Computational Exercise (Simulation)

Duration: 3 hours · **Level:** Intermediate → Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Lab](#) · → [Next Lab](#)

■ [Related Module: Module 11 — Resistance Evolution](#) · ■ [Dataset: Exercise 3 — Resistance frequencies](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Implement a simple single-locus selection recursion for a resistance allele under Bt exposure with and without refuges.
2. Explain why resistance build-up follows a slow–fast (S-shaped) trajectory.
3. Quantify the refuge effect: years to resistance threshold at 0%, 10%, 20% refuge.
4. Identify the two parameters to which the model outcome is most sensitive — and why they are hard to estimate.

Background

Bt crops impose intense, season-long selection on target pests. The **high-dose/refuge strategy** exploits population genetics: refuges of non-Bt hosts sustain susceptible alleles that dilute resistance when rare. This lab makes the strategy quantitative (Module 11 provides the biology; Module 25.3 the modelling context).

Scientific principle

Single-locus recursion with resistance allele frequency q , relative fitnesses on Bt crop w_{RR} , w_{RS} , w_{SS} (resistant homozygotes survive; heterozygotes survive only if the dose is not "high"), and refuge fraction r :

$$q' = [q^2 \cdot w_{RR} + q(1-q) \cdot w_{RS}] \cdot r + q_{on_bt} \cdot (1-r) \quad (\text{normalised})$$

with q_{on_bt} computed from Bt-field fitnesses and r the refuge share. Two features drive the classic S-curve: (i) selection on a **rare recessive** allele is weak (heterozygotes die on Bt), so early growth is slow; (ii) once q is high, most resistance alleles sit in protected homozygotes and frequency rockets.

Materials/data

- DATA/resistance/ — simulated resistance-frequency trajectories over 25 generations at refuge levels 0%, 10%, 20% (seeded). **Simulated data.**
- Python 3 (numpy, pandas, matplotlib); reference: DATA/analysis_demo.py Exercise 3.
- Provide your own spreadsheet alternative if Python access is limited.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Run the baseline** recursion ($r=0$) and plot q vs generation.
2. **Add refuges** ($r=0.1, 0.2$) and overlay the three trajectories.
3. **Threshold analysis:** generations until $q = 0.5$ for each refuge level (Table 1).
4. **Sensitivity:** vary initial q (0.0001, 0.001, 0.01) and resistance dominance h (0, 0.1, 0.5); record years-to-resistance in Table 2.
5. **Management reading:** convert each result into one risk-management sentence (Module 18 framing).

6. **Challenge:** add fitness costs ($w_{RR} = 1 - c$ on refuge) and find the cost c that stabilises q under 10% refuge.

Data tables

Table 1 — refuge effect (from provided simulation):

Refuge (%)	Generation q reaches 0.5	Management sentence
0		
10		
20		

Table 2 — sensitivity:

q	h	Years to $q=0.5$ (10% refuge)
0.0001	0.0	
0.001	0.1	
0.01	0.5	

Calculations

- Implement the recursion (course `analysis_demo.py` shows a reference implementation; students must write their own and compare outputs — identical results validate both).
- Report the resistance **growth rate** $\Delta q/q$ per generation for generations 1–5, 10–15, 20–25; relate the pattern to the S-curve.

Expected results

(Simulated — documented by `analysis_demo.py`.) With no refuge, resistance rises to fixation within the simulation window; 10% refuge delays it several-fold; 20% refuge pushes it beyond the horizon. Slow early growth while q is recessive-rare, explosive growth once q is common — the documented nonlinear build-up with a dramatic refuge effect.

Interpretation

- The refuge works **only while resistance is rare** — monitoring (Module 19) is what tells you the strategy is still working; by the time field failures appear, q is already high.
- Sensitivity results explain why regulators demand conservative assumptions on initial frequency and dominance: model output is dominated by parameters measured with difficulty.
- Resistance management is not a static rule; it is adaptive management with monitoring triggers (Module 19 workflow).

Troubleshooting

Symptom	Cause	Fix
q hits 1.0 with NaN afterwards	No normalisation	Normalise mean fitness each generation
No S-shape, straight line	Dominance set high ($h \approx 0.5$)	That is the point — re-run with $h=0$ and compare
Refuges look useless	Refuge insects also on Bt (spatial error)	Refuge fraction must apply to <i>non-Bt</i> mating pool

Questions

1. Why is early selection weak when resistance is recessive, even under a "high-dose" regime?
2. Your 20% refuge delays resistance beyond the horizon. Does that prove resistance will never evolve? What is the honest statement?
3. How would stacked pyramids (two toxins) change the effective dominance?
4. What monitoring design detects $q=0.001$? Is that feasible?

Viva questions

1. Define the high-dose/refuge strategy.
2. Why must refuges produce **susceptible adults that mate with in-field adults**?
3. What is cross-resistance and why do pyramids reduce it?
4. Name two real-world factors the simple model omits.

Instructor answer key

- Q1: Almost all resistance alleles sit in heterozygotes, which die on a high-dose Bt crop; selection acts only on the tiny RR fraction, so $\Delta q \propto q^2$ is very small while q is rare.
- Q2: No — delay beyond the horizon is not absence; with longer horizons or lower fitness costs, evolution continues; honest statement: "under these assumptions, q stays below 0.5 for >25 generations."
- Q3: With two toxins and no cross-resistance, insects heterozygous at both loci die, effectively reducing functional dominance and slowing evolution (the pyramid rationale).
- Q4: Sampling for $q=0.001$ requires $\approx 3,000$ alleles per population for $\sim 95\%$ detection chance; feasible only with pooled/sequencing methods and dense coverage.

Advanced challenge

Replace the deterministic recursion with a stochastic one (binomial sampling of alleles each generation, $N=10,000$) and run 1,000 replicates per refuge level. Report the **distribution** (not the mean) of years-to-resistance and the probability of resistance within 10 years. How does the stochastic answer change the management sentence from Table 1?

Previous: [Lab 04 — Non-Target Data Analysis](Lab-04-Non-Target-Data-Analysis.md) · Next: [Lab 06 — Environmental Exposure Analysis](Lab-06-Environmental-Exposure-Analysis.md)

Lab 06 — Environmental Exposure Analysis

Activity type: Computational Exercise

Duration: 2.5 hours · **Level:** Intermediate

Navigation: ↑ [Course Home](#) · ← [Previous Lab](#) · → [Next Lab](#)

■ [Related Module: Module 13 — Environmental Fate and Exposure](#) · ■ [Dataset: Exercise 7 — Exposure concentrations](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Map an exposure pathway from source to receptor (crop → pollen → plant → herbivore → predator).
2. Quantify exposure magnitude at each pathway step from simulated field data.
3. Combine concentration × duration × rate into an exposure estimate with stated assumptions.
4. Identify where in a pathway exposure is controlled by biology vs by management.

Background

Risk = hazard × exposure (conceptually). Module 13 teaches the pathway logic; this lab makes one full pathway quantitative: **Bt maize** → **pollen** → **drift onto non-target host plants** → **herbivore larvae feeding** → **predator consuming larvae**. Each step multiplies (and usually attenuates) exposure; each step is also a place where monitoring or management can intervene.

Scientific principle

Step-wise exposure propagation:

```
E_step = C_source × Transfer(step) × Attenuation(step)
E_total = ∏ steps
```

Transfer factors (pollen drift fraction, deposition density, larval consumption rate) come from the dataset; attenuation (protein degradation, metabolic dilution) comes from the fate data. The final receptor dose is typically **orders of magnitude below** the source concentration — quantifying that attenuation chain *is* the assessment.

Materials/data

- DATA/exposure/ — simulated: pollen protein concentration, pollen densities on sticky stations at 0/5/25/100 m into the margin, protein degradation over days, larval consumption rates, predator diet composition. **Simulated data.**
- Python 3; reference: DATA/analysis_demo.py Exercise 7.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Draw the pathway** (five boxes, arrows labelled with the transfer factor each represents).
2. **Pollen density by distance** — mean and CI per station distance; plot decay curve.
3. **Protein decay** — estimate the degradation half-life from the time series; fit exponential.
4. **Multiply the chain** — compute predator-reachable dose under (a) field-edge, (b) 25 m, (c) 100 m scenarios.

5. **Compare doses** with the species' high-dose laboratory threshold (provided in the dataset notes) → simple hazard quotient per scenario.
6. **Interpret** — which step controls the final dose; where would management act?
7. **Challenge**: add flowering-synchrony variability (± 10 days) and propagate it through the chain (Exercise 10 methods).

Data tables

Step	Transfer/attenuation factor	Value (mean)	95% CI	Source of variability
Pollen protein conc.	—			
Deposition at 5 m	fraction of source			
Deposition at 25 m	fraction			
Deposition at 100 m	fraction			
Protein half-life	days			
Larval consumption	$\mu\text{g/day}$			
Predator dose	$\mu\text{g/day}$			

Calculations

- Exponential decay fit for both distance (pollen) and time (protein).
- Hazard quotient $HQ = \text{receptor dose} / \text{high-dose threshold per scenario}$; interpret $HQ < 1$ vs > 1 with its uncertainty.
- Propagate one parameter's CI through the chain (worst-case vs best-case arithmetic).

Expected results

(Simulated — documented by `analysis_demo.py`.) Pollen deposition falls steeply with distance; protein decays with a half-life of a few days; the receptor dose at field edge remains 10–100× below the high-dose threshold, and at 100 m the margin widens further. The chain demonstrates **exposure attenuation by orders of magnitude** — the quantitative backbone of the monarch-case logic (Case 2).

Interpretation

- The **biology-controlled** steps (pollen shed, degradation) set the ceiling; the **management-controlled** steps (border rows, distance, timing vs flowering) offer the levers.
- An HQ near 1 under worst-case assumptions flags the parameter to measure better — not necessarily a risk (Module 16–17 chain: difference → relevance → adverse? → exposure? → risk?).
- Every number in the chain carries uncertainty; the honest product is an interval, not a point (Module 23).

Troubleshooting

Symptom	Cause	Fix
HQ enormous	Threshold misread (per-day vs per-larva units)	Check units at every step — the classic error
Decay fit poor	Tail measured imprecisely	Fit on log scale; report residuals
Dose exceeds source	Transfer factors > 1 (unit error)	Transfer fractions must be ≤ 1 ; audit the table

Questions

1. Which single parameter most influences the final predator dose, and how do you know?
2. Why does the hazard quotient use the *high-dose laboratory* threshold rather than a field observation?
3. Name two management interventions and the pathway step each targets.
4. Your worst-case HQ = 0.8. What is the correct risk statement?

Viva questions

1. Define exposure pathway and name its components for Bt pollen.
2. Why does attenuation across a trophic step not always reduce risk proportionally?
3. What is flowering synchrony and why does it gate the whole pathway?
4. Distinguish environmental fate from exposure.

Instructor answer key

- Q1: Usually the deposition-at-edge fraction (largest relative CI and leverage); establish via the sensitivity calculation (Step 7 / Exercise 10).
- Q2: A conservative screening device — if even the laboratory ceiling dose is not approached, pathway risk is negligible for that route; it is a bound, not a field-relevant estimate.
- Q3: Border rows / isolation distance target the deposition step; planting-date choice targets the synchrony step; residue-timing targets the decay step.
- Q4: "Under worst-case assumptions the receptor dose remains below the laboratory high-dose threshold, but the margin is narrow (HQ 0.8); measurement of the dominant parameter is warranted before concluding negligible risk." — not "safe," not "dangerous."

Advanced challenge

Build the full pathway as a Monte Carlo (1,000 draws from each parameter's distribution, Exercise 10 tooling) and report the distribution of HQ at field edge and 25 m. What percentage of runs exceed HQ = 1, and which parameters drive the exceedance tail?

Previous: [Lab 05 — Resistance Evolution Simulation](Lab-05-Resistance-Evolution-Simulation.md) · Next: [Lab 07 — Compositional Assessment](Lab-07-Compositional-Assessment.md)

Lab 07 — Compositional Assessment Data Interpretation

Activity type: Computational Exercise / Case Study

Duration: 2.5 hours · **Level:** Intermediate

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Next Lab](#)

■ **Related Modules:** [Module 5 — Phenotypic and Compositional Assessment](#), [Module 14 — Food and Feed Safety](#) · ■

[Dataset: Exercise 6 — Composition](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Interpret compositional (proximate + key nutrients/anti-nutrients) comparisons between a GM line and its conventional comparator.
2. Apply equivalence-interval logic rather than raw significance testing.
3. Recognise the role of natural range data from conventional varieties (site-to-site, year-to-year variation).
4. Explain what compositional equivalence can and cannot establish (the bridge to Module 16's unintended-effects logic).

Background

Food/feed safety assessment (Module 14) begins with **comparative composition**: is the GM crop materially equivalent to its conventional counterpart, allowing for the intended change? The comparator is not "perfectly identical" — conventional crops vary hugely with site, year, and genotype. Regulators therefore use reference ranges built from many conventional lines. This lab trains the reading of such tables — the single most common dossier artefact students will meet in practice.

Scientific principle

For each analyte:

```
Difference = GM mean - Comparator mean
95% CI on difference (or equivalence test against pre-set margins)
```

Then place the difference in context: is GM mean **within the conventional reference range**? An equivalence margin (e.g., $\pm 20\%$ of comparator mean, as used illustratively in regulatory guidance) plus a confidence-interval test (TOST: two one-sided tests) answers "equivalent" properly; a plain t-test only answers "detectably different," which low power can hide and high power always finds.

Materials/data

- DATA/composition/ — simulated proximates (protein, fat, fibre, ash, moisture), minerals, vitamins, anti-nutrients for GM line, near-isogenic comparator, and six conventional reference lines across sites.

Simulated data.

- Python 3; reference: DATA/analysis_demo.py Exercise 6.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Compute** GM vs comparator means, % difference, 95% CI for each analyte (Table 1).
2. **Equivalence test** at $\pm 20\%$ margins — classify each analyte equivalent / not equivalent / inconclusive.

3. **Reference-range check** — is the GM mean inside the pooled conventional range (min–max of the six reference lines)?
4. **Flag intended differences** (e.g., the trait's expected metabolic effect) and separate them from *unintended* flags.
5. **Interpret** — write a dossier-style conclusion paragraph for the compositional section.
6. **Challenge**: re-run the equivalence test with $\pm 10\%$ margins; how do conclusions move and what does that teach about margin choice?

Data tables

Table 1 (per analyte):

Analyte	Comparator mean	GM mean	% diff	95% CI of diff	Within conventional range?	Verdict
Protein						
Fat						
Fibre						
(...all analytes)						

Table 2 — summary counts: equivalent / not equivalent / inconclusive / intended-difference.

Calculations

- Standard equivalence test (TOST) at the stated margin; show one worked analyte by hand.
- Pooled conventional reference range (min–max across lines/sites).
- Optional: mixed-model site effect to show why site, not GM status, often dominates variance.

Expected results

(Simulated — documented by `analysis_demo.py`.) Most analytes are equivalent and within reference ranges; one analyte shows a small shift consistent with the intended metabolic change; one or two fall "inconclusive" at small margins — demonstrating that margin choice drives verdicts and must be pre-declared, not post-hoc.

Interpretation

- **Equivalent $\neq$ identical:** compositional assessment supports "no indication of unexpected compositional change," not "proven safe."
- **Intended differences are expected:** e.g., a provitamin-A biofortified line *must* differ in carotenoids; the assessment asks whether *other* analytes are unremarkable.
- **A significant p-value on one analyte among fifty is expected by chance alone** — multiplicity logic (Module 24); look for consistency across sites and the pattern across related analytes before flagging.

Troubleshooting

Symptom	Cause	Fix
Half the analytes "not equivalent"	Comparing to the wrong comparator (not near-isogenic, different maturity)	Verify comparator identity; check agronomic confounders
CI of difference absurdly wide	Too few sites/replicates	Report power limitation honestly; do not claim equivalence
One extreme outlier analyte	Unit/conversion error (fresh vs dry weight)	Normalise basis first; recompute

Questions

1. Why is a plain t-test the wrong tool for "is this crop compositionally equivalent?"
2. One analyte differs by 12% (CI 6–18%), inside the conventional range. Flag or not — and what further evidence would you want?
3. Why must the equivalence margin be pre-declared?
4. What can compositional analysis *never* detect?

Viva questions

1. Define substantial equivalence and state one criticism of it.
2. What is a near-isogenic comparator and why does maturity matter?
3. Name three analyte classes always assessed in a crop dossier.
4. How do anti-nutrients figure in feed-safety assessment?

Instructor answer key

- Q1: Equivalence is a claim about being *similar within margins*; a t-test tests difference and conflates power with equivalence — underpowered tests "pass" spuriously; TOST targets the right hypothesis.
- Q2: Within-range differences are typically not flagged alone; follow-up would seek consistency across sites/years and a plausible mechanism (or absence of one) before classification.
- Q3: Post-hoc margins invite verdict-shopping; pre-declaration keeps the equivalence claim falsifiable and comparable across dossiers.
- Q4: Compounds not measured, unknown modes of action, downstream product effects (processing), long-term dietary interactions — composition is one line of evidence, never the whole.

Advanced challenge

Simulate a 50-analyte panel with no true GM effect and show that at $\alpha=0.05$ roughly 2–3 analytes appear "significantly different" by chance. Then show how the reference-range check suppresses these false flags. Write the paragraph a risk assessor should include about multiplicity.

Previous: [Lab 06 — Environmental Exposure Analysis](Lab-06-Environmental-Exposure-Analysis.md) ·

Next: [Lab 08 — Dose-Response Analysis](Lab-08-Dose-Response-Analysis.md)

Lab 08 — Dose–Response Data Analysis

Activity type: Computational Exercise

Duration: 2.5 hours · **Level:** Intermediate → Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Lab](#) · → [Next Lab](#)

■ [Related Modules: Module 15 — Allergenicity and Toxicity Assessment, Module 8 — Non-Target Organisms](#) · ■ [Dataset: Exercise 8 — Dose-response](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Fit and interpret a logistic dose–response model; define and estimate LC50 conceptually.
2. Contrast dose–response curves of a target pest and a non-target species exposed to the same protein.
3. Connect curve parameters to ERA statements (specificity, margins of exposure).
4. Explain what dose–response evidence can and cannot conclude about chronic/sublethal effects.

Background

Toxicological assessment (Module 15) rests on dose–response: response increases with dose in a defined way, and *specificity* is observed as curve separation between sensitive (target) and insensitive (non-target) organisms. For Bt proteins the target/non-target separation is the scientific basis for environmental selectivity; this lab quantifies that logic on simulated data.

Scientific principle

Logistic model for mortality at dose d :

$$p(d) = 1 / (1 + \exp(-\beta \cdot (\ln d - \ln \text{LC50})))$$

- **LC50** — dose killing 50% of test organisms (a hazard-characterisation parameter).
- **Slope β** — steepness; shallow slopes complicate threshold-setting.
- **Separation of LC50s** between species = selectivity margin; the ERA uses the margin relative to *realised environmental exposure* (Lab 06 logic).

Materials/data

- DATA/dose-response/ — simulated mortality at 6–8 doses for: (i) target lepidopteran larva, (ii) non-target predator, (iii) non-target soil Collembolan; $n=50$ per dose; **simulated data** with a documented LC50 separation of several orders of magnitude.
- Python 3; reference: DATA/analysis_demo.py Exercise 8.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Plot** raw mortality vs dose (log x) for the three species on one figure.
2. **Fit** logistic curves (or read from the reference implementation); record LC50 and slope per species (Table 1).
3. **Compute** the selectivity margin $\text{LC50}(\text{non-target})/\text{LC50}(\text{target})$.
4. **Overlay** field-realistic exposure estimates (from Lab 06 dataset notes) as vertical lines; compute margins of exposure.
5. **Interpret** the three-species picture as a risk-assessment paragraph.

6. **Challenge:** simulate sublethal endpoints (development time) and discuss why curve-based LC50 logic does not automatically cover them.

Data tables

Table 1:

Species	LC50 (est.)	Slope β	95% CI of LC50
Target larva			
Non-target predator			
Non-target Collembolan			

Table 2 — margins of exposure at the three Lab-06 scenarios: species × scenario grid; each cell = LC50 / estimated field dose.

Calculations

- LC50 by interpolation on the fitted curve (state method).
- Selectivity margins (show one division with units).
- Margin of exposure per scenario (state conservative vs central estimates).

Expected results

(Simulated — documented by `analysis_demo.py`.) Target species: steep curve, low LC50. Non-targets: high LC50s — several orders of magnitude above the target's — with shallow curves. Field-realistic exposure lines sit far below every non-target LC50. The documented conclusion: strong selectivity margin; risk to these non-targets via this route negligible **for acute lethality at these exposures**.

Interpretation

- The *combination* of steep target response, high non-target LC50s, and low environmental exposure is what makes the environmental conclusion robust — remove any one leg and the conclusion weakens (good illustration of weight of evidence).
- Chronic and sublethal endpoints (development, fecundity, behaviour) are **not** captured by acute LC50 logic; Module 15's weight-of-evidence discussion and Lab 04's tiered design address them. Do not over-claim from this lab.
- Slope matters for thresholds: with shallow slopes, small concentration errors move the predicted response a lot — one more reason to propagate uncertainty.

Troubleshooting

Symptom	Cause	Fix
Fit fails at extremes	0% / 100% mortality anchors	Use a link that handles binomial extremes, or trim anchor doses for fitting while keeping them in plots
LC50 outside dose range	Weak dose placement	Note as a design limitation; bracket the LC50 properly next time
Species curves cross	Interaction of slope and scale	Discuss in terms of low-dose behaviour, not just LC50

Questions

1. Why is a several-orders-of-magnitude selectivity margin more convincing than a factor of two?

2. A non-target LC50 cannot be reached at any tested dose. How do you report an LC50 you could not estimate?
3. Your field-exposure line sits below the target species' LC10. What does that say about *target* mortality expectations?
4. Why doesn't this lab's conclusion extend to chronic effects?

Viva questions

1. Define dose–response and LC50.
2. What is a margin of exposure and how is it used?
3. Why do Bt proteins require midgut activation — and what does that imply for specificity?
4. Name two sublethal endpoints relevant to non-target risk.

Instructor answer key

- Q1: Large margins buffer every uncertainty in the chain (exposure, slope, species extrapolation); a factor of two can be erased by a single measurement error, orders of magnitude cannot.
- Q2: Report as "> highest tested dose" with the value stated — an inequality, not a number; that is standard hazard-characterisation practice.
- Q3: It predicts little target mortality in the field — a paradox typical of forced-dose comparisons and a cue to check exposure assumptions (in the crop itself, target larvae feed on tissue, not pollen — exposure route differs).
- Q4: Acute curves measure lethality over a fixed exposure window; sublethal/chronic effects need life-table or multigeneration designs with their own dose-response logic.

Advanced challenge

Fit the curves with a bootstrap over organisms (resample within dose) and report the *distribution* of the selectivity margin. What is the 5th percentile of the margin, and does the ERA conclusion survive at that conservative percentile?

Previous: [Lab 07 — Compositional Assessment](Lab-07-Compositional-Assessment.md) · Next: [Lab 09 — Risk-Ranking Matrix](Lab-09-Risk-Ranking-Matrix.md)

Lab 09 — Risk-Ranking Matrix

Activity type: Computational Exercise / Case Study

Duration: 2 hours · **Level:** Intermediate

Navigation: ↑ [Course Home](#) · ← [Previous Lab](#) · → [Next Lab](#)

■ [Related Modules: Module 17 — Risk Characterization, Module 2 — Hazard, Risk and Exposure](#) · ■ [Dataset: Exercise 9 — Risk-matrix scores](#) · ■ [Assessment: Case-Based Questions](#)

Learning objectives

1. Construct a qualitative risk matrix (likelihood × consequence) for identified GMO hazards.
2. Rank hazards consistently and justify placements with evidence, not intuition.
3. Demonstrate matrix sensitivity: how placements shift with scoring definitions.
4. Connect matrix output to management prioritisation and monitoring design.

Background

Risk characterization (Module 17) often begins qualitatively: list hazards, estimate likelihood and consequence, rank. A risk matrix is a communication and prioritisation tool — not a measurement. Used carelessly it manufactures false precision; used well it makes the reasoning transparent and contestable. This lab teaches the disciplined use.

Scientific principle

Risk score = Likelihood score × Consequence score (ordinal scales, e.g., 1–5)

Each axis is anchored with **written descriptors** (what makes a "3" a 3?), and every placement must cite the evidence used. Matrix cells map to action classes (accept / manage / monitor / reject). Sensitivity analysis asks: if two reasonable assessors score the same hazard, do they land in the same action class? If not, the *descriptors*, not the assessors, need work.

Materials/data

- DATA/risk-matrix/ — simulated scored hazard/exposure pairs from a hypothetical Bt-cotton ERA (hazard list with likelihood/consequence scores and one-line justifications). **Simulated data.**
- Hazard list from Lab 01's worksheet (your own version).
- Python (matplotlib heatmap) or paper grid.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Define scales** — write anchor descriptors for likelihood 1–5 and consequence 1–5 *before* scoring anything (Table 1).
2. **Score** the course's standard hazard list (below) using those anchors; cite evidence per score.
3. **Plot** the 5×5 matrix with hazards placed; colour action classes.
4. **Cross-validate** — swap matrices with another group; identify placements where descriptors allowed divergence.
5. **Refine** descriptors that produced divergence; re-score; record placement changes.

6. **Link to management** — for the top three hazards, name the management/monitoring instrument each would trigger (Modules 18–19).

7. **Challenge:** replace multiplicative scoring with an ordinal 3×3 matrix; which hazards move action class?

Data tables

Table 1 — scale anchors (write your own):

Score	Likelihood anchor	Consequence anchor
1		
2		
3		
4		
5		

Table 2 — hazard scores:

Hazard	L	C	Score	Action class	Evidence cited
Resistance evolution (Cry1Ac)					
Gene flow to wild relatives					
Non-target predator effects					
Soil-community shift					
Volunteer persistence					
Horizontal gene transfer to microbes					
Herbicide-management shift					
Food/feed safety (new protein)					

Calculations

- Score = $L \times C$; action-class map: <4 accept/monitor · 4–9 manage · 10–16 manage intensively · ≥ 17 reject/re-design (illustrative mapping — declare your own).

Expected results

(Simulated dataset — documented by `analysis_demo.py`.) Resistance evolution scores highest (high likelihood under continuous selection; high consequence for trait durability); food/feed protein hazard scores low on likelihood given weight-of-evidence but non-trivial consequence; HGT-to-microbes scores low on consequence under current evidence. Cross-group placements will typically diverge by one cell on 2–3 hazards — that finding *is* the lesson about descriptor quality.

Interpretation

- Matrices make prioritisation **explicit and contestable** — their value; they do not measure risk — their limit.

- Placement disputes almost always trace to likelihood anchors (rare-event calibration) — write those anchors with the most care.
- Every high-ranked hazard should map to a management or monitoring line in the ERA report; a top-ranked hazard with no instrument is an unmanaged risk, and the assessment must say so.

Troubleshooting

Symptom	Cause	Fix
Everything lands mid-matrix	Averaging the axes ("3ish, 3ish")	Force extremity: anchors must describe <i>observable</i> endpoints
Scores drift between groups	Vague anchors	Rewrite as frequencies/magnitudes, not adjectives
Matrix used as final output	Tool misuse	Attach the evidence table; matrix is presentation, assessment is the reasoning

Questions

1. Why should anchors be written before scoring?
2. Two groups place the same hazard one cell apart. What — precisely — resolved it?
3. Which hazard in Table 2 has the largest gap between scientific evidence and matrix placement, and why?
4. Why is HGT-to-microbes scored low on consequence despite being famous in public debate?

Viva questions

1. Distinguish qualitative, semi-quantitative, and quantitative risk characterization.
2. What makes a matrix cell "actionable"?
3. Why can a low-likelihood hazard still be the top priority?
4. How does the matrix connect to adaptive management?

Instructor answer key

- Q1: Pre-declared anchors prevent post-hoc fitting of scores to preferred conclusions — the matrix equivalent of pre-registering hypotheses.
- Q2: One or more anchor descriptors were ambiguous; the fix is descriptor language, not scorer calibration (calibration invites groupthink).
- Q3: Typically HGT — public salience high, assessed consequence low (Module 9's evidence review); teaches the communication/assessment gap (Module 21).
- Q4: Under current evidence natural transformation of soil bacteria with plant transgenes is exceedingly rare and without demonstrated establishment — consequence is scored on evidence, not fame.

Advanced challenge

Score all eight hazards twice: once with *your* evidence base, once deliberately withholding the comparator studies (sprayed-conventional baselines). Compare matrices. Write a short note on how comparator availability — not logic — changes risk rankings, and what that implies for regulatory data requirements.

Previous: [Lab 08 — Dose-Response Analysis](Lab-08-Dose-Response-Analysis.md) · Next: [Lab 10 — Uncertainty and Sensitivity Analysis](Lab-10-Uncertainty-Sensitivity-Analysis.md)

Lab 10 — Uncertainty and Sensitivity Analysis

Activity type: Computational Exercise

Duration: 3 hours · **Level:** Advanced

Navigation: ↑ [Course Home](#) · ← [Previous Lab](#) · → [Next Lab](#)

■ [Related Modules: Module 16 — Unintended Effects and Uncertainty, Module 17 — Risk Characterization](#) · ■ [Dataset: Exercise 10 — Uncertainty](#) · ■ [Assessment: Data-Interpretation Questions](#)

Learning objectives

1. Propagate parameter uncertainty through a risk-relevant model with Monte Carlo simulation.
2. Separate variability (real-world heterogeneity) from uncertainty (state of knowledge).
3. Rank input parameters by influence on the output (sensitivity analysis) and justify resource allocation for further measurement.
4. Communicate an interval-based conclusion without false precision.

Background

Every estimate in an ERA — outcrossing frequency, LC50, exposure dose — carries uncertainty. Module 16 teaches the taxonomy; this lab runs the machinery. The model: an **outcrossing-estimate** model combining edge frequency, decay constant, field area, and flowering overlap into a field-scale transgene-flow estimate. Students propagate uncertainty from the four inputs and discover which input dominates — the result changes what they would measure next.

Scientific principle

Monte Carlo: draw each input from its stated distribution N times; compute the output each time; the output distribution is the uncertainty statement.

$$F = a \cdot \exp(-b \cdot d) \cdot S \cdot O \text{ (edge freq, decay, area factor, overlap factor)}$$

Sensitivity: correlation (or rank-correlation) of each input with the output across draws — a practical proxy for "which uncertainty matters."

Materials/data

- DATA/uncertainty/ — simulated parameter distributions (mean, CI, shape) for the four model inputs, plus 10,000-draw output summary. **Simulated data.**
- Python 3 (numpy); reference: DATA/analysis_demo.py Exercise 10.
- Output from Lab 03 (your fitted a , b) may be substituted for the defaults.

Safety

No biosafety concerns.

Step-by-step workflow

1. **Define the model** on paper — inputs, units, plausible ranges (Table 1).
2. **Set distributions** — uniform (sparse knowledge) vs normal (measured) vs lognormal (positive quantities); justify each choice.
3. **Run** 10,000 draws; plot the output distribution (histogram + 90% interval).
4. **Sensitivity** — rank inputs by rank-correlation with output (Table 2).
5. **Tornado plot** — effect of moving each input from its low to high bound, one at a time.

6. **Variability vs uncertainty** — treat flowering overlap as *variability* (site-years) and edge frequency as *uncertainty* (measurement); report the two intervals separately.

7. **Challenge:** add correlation between two inputs (a and b are often anti-correlated in fits) and show the effect on the output interval.

Data tables

Table 1 — input definitions:

Input	Baseline	Distribution + justification	Source
Edge frequency a			
Decay constant b			
Field-area factor S			
Overlap factor O			

Table 2 — sensitivity output:

Rank	Input	Rank-correlation with output	Action: measure better?
1			
2			
3			
4			

Calculations

- 10,000-draw Monte Carlo; report median, 5th, 95th percentiles.
- Rank (Spearman) correlations; one-line interpretation each.
- Interval comparison: deterministic (best-estimate inputs) vs probabilistic interval — quantify the gap false precision was hiding.

Expected results

(Simulated — documented by `analysis_demo.py`.) The deterministic point estimate sits inside a wide 90% interval; **isolation-distance/decay parameter (b) dominates the output** (the documented interpretation: isolation distance dominates, not edge outcrossing). Acting on the ranking: the next field campaign should measure distance-decay more precisely, not the near-field frequency — a concrete, defensible resource-allocation statement.

Interpretation

- The ranking converts uncertainty from a paragraph of hedging into a **decision about what to measure** — this is the practical payoff of sensitivity analysis.
- An interval that crosses a management threshold is *not* an excuse for inaction: it triggers either targeted measurement or conservative management (Module 19 adaptive logic).
- Reporting format that earns marks: "Median field-scale flow 0.12% (90% interval 0.03–0.45%), dominated by uncertainty in the distance-decay parameter; the median exceeds the 0.1% threshold in X% of draws, driven principally by parameter b."

Troubleshooting

Symptom	Cause	Fix
Interval implausibly narrow	Inputs treated as exact	Every input needs a distribution — audit Table 1
Output distribution skewed wildly	Lognormal inputs with large σ	Check parameter units and tails; plot input draws
Sensitivity ranks unstable	Too few draws	Increase N; use rank correlation (robust to monotone noise)

Questions

1. Why is rank correlation preferred over Pearson for sensitivity here?
2. Your 90% interval spans a management threshold. Give three defensible responses and the assumptions each carries.
3. What does it mean that parameter *b* dominates — in terms of the *biology* of pollen flow?
4. Why report variability and uncertainty separately?

Viva questions

1. Define measurement, model, and sampling uncertainty with an ERA example each.
2. What is a tornado plot used for?
3. Distinguish worst-case, plausible-worst-case, and central estimates.
4. Why is "conservative" not automatically "better" in risk assessment?

Instructor answer key

- Q1: Relationships with the output may be monotone but non-linear; rank correlation captures influence without linearity assumptions and is outlier-robust.
- Q2: (i) Measure the dominant parameter better (costs time, buys precision); (ii) manage conservatively now (acts as if near-worst case); (iii) redesign the option/decision to be robust across the interval (robustness framing). Each embeds a different attitude to the cost of error.
- Q3: Decay with distance is the strongest lever on *total* field-scale flow — it aggregates every far-field receptor; edge frequency, by contrast, affects only a thin band.
- Q4: They answer different questions and demand different remedies — variability needs spatial/temporal design (more sites/years), uncertainty needs better measurement; conflating them misdirects both.

Advanced challenge

Replace the fixed distributions with a two-level analysis: *within-model* Monte Carlo (as above) plus *between-model* comparison (exponential vs power-law decay from Lab 03). Report the model-uncertainty contribution to the total interval — the honest way to present structural uncertainty in a regulatory summary.

Previous: [Lab 09 — Risk-Ranking Matrix](Lab-09-Risk-Ranking-Matrix.md) · Next: [Lab 11 — Case-Study Regulatory Assessment](Lab-11-Case-Study-Regulatory-Assessment.md)

Lab 11 — Case-Study Regulatory Assessment

Activity type: Case Study (role-play simulation)

Duration: 3 hours · **Level:** Advanced

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Next Lab](#)

■ [Related Modules: Module 22 — Regulatory Frameworks, Module 23 — Case Studies](#) · ■ [Assessment: Case-Based Questions](#) · ■ [Cheat Sheet](#)

Learning objectives

1. Apply the full ERA workflow (Module 3) to a novel dossier under time constraints.
2. Separate scientific assessment outputs from regulatory decision inputs — and experience the tension.
3. Draft the core sections of a risk-assessment report: problem formulation, characterization, uncertainty statement.
4. Defend an evidence-based position against informed counter-argument.

Background

Modules 22–23 showed that jurisdictions reach different decisions on similar evidence. This lab makes students *live* that process: a fictional-but-realistic dossier is assessed by competing roles (assessor, developer, public-interest reviewer, regulator), each bound by the same evidence. The outcome is not scripted — and the debrief examines why.

Scientific principle

The assessment/decision boundary (Module 22):

- **Assessors** characterise risk and uncertainty with no decision authority.
- **Regulators** set acceptable-risk thresholds, conditions, and burdens of proof — policy-laden choices.
- **Stakeholders** contribute values and local knowledge — legitimate inputs to *decision*, not to *characterization*.

Role instructions enforce this separation; violations are graded.

Materials/data

- **Dossier (provided by instructor or generated from the course):** a fictional stacked-trait Bt + herbicide-tolerant (HT) maize proposed for release in a region with irrigated maize, smallholder refuges unlikely to be enforced, and a sexually compatible wild relative present. Dossier includes: molecular characterization summary, compositional table (Lab 07 style), three non-target studies (Lab 04 style), gene-flow data (Lab 03 style), resistance model outputs (Lab 05 style), and a developer-proposed monitoring plan.
- Role cards (below), the 12-section template from Module 23, course Cheat Sheet.
- No real dossiers or proprietary data are used.

Safety

No biosafety concerns (paper exercise). Professional-integrity rule: no role may invent data; uncertainty must be stated, not papered over.

Step-by-step workflow

1. **Formation (10 min).** Groups of 4: Assessor (A), Developer (D), Public-interest reviewer (P), Regulator (R).
2. **Independent reading (25 min).** Each reads the dossier and lists: strongest evidence, weakest evidence, missing studies.
3. **Assessor stage (30 min).** A drafts problem formulation + hazard list; D and P each submit their top three concerns; A integrates (no decisions allowed).
4. **Characterization stage (30 min).** Classrooms' Lab 03/05/09 outputs are cited as the quantitative backbone; A produces the risk-characterization table with uncertainty flags.
5. **Decision stage (30 min).** R receives: A's characterization, D's proposed conditions, P's objections. R drafts a decision with conditions and a monitoring trigger table.
6. **Public hearing (20 min).** 3-minute statements from D, P, A; R answers using only the record.
7. **Debrief (25 min).** Compare decisions across groups; map divergences to assessment differences vs value differences.

Data tables

Characterization table (Stage 4):

Hazard	Likelihood	Consequence	Uncertainty flag	Basis (dossier section)

Decision record (Stage 5):

Element	Regulator's entry
Decision	
Conditions	
Monitoring triggers	
Review date	
Residual uncertainties acknowledged	

Calculations

None new — cite Lab 03/05/07/08/09 results as the dossier's quantitative layer.

Expected results

Groups typically converge on the hazard list (the science is shared) but diverge on: refuge enforcement as a condition, the acceptability of monitoring-triggered withdrawal, and the standard of proof for the wild-relative pathway — precisely the assessment/decision boundary the lab exists to teach.

Interpretation

- Divergence is not failure: the debrief question is *which* divergence came from evidence and *which* from values. Both are legitimate; only one belongs in the characterization table.
- A decision with conditions is not a "weaker yes" — conditions (refuges, monitoring triggers, review dates) are the regulator's principal risk-management instruments (Module 18).
- The reviewer's role is not to oppose but to test whether the characterization's uncertainties are honestly carried into the decision record.

Troubleshooting

Symptom	Cause	Fix
Regulator rubber-stamps developer	Role capture	Require R to cite one dossier weakness and one condition responding to it
Assessor makes the decision	Boundary violation	Instructor intervention; re-issue role cards
Debate loops on "is GMO safe"	Framing collapse	Return to the hazard table; "safe" is not an assessable claim (Lab 01)

Questions

1. Which dossier weakness did the developer have the strongest incentive to minimise, and did the process surface it?
2. What condition could the regulator set that does not depend on future grower compliance?
3. Where did your group's decision rest on values rather than evidence? Name it explicitly.
4. Which monitoring trigger, if pre-agreed, would most change the risk profile itself?

Viva questions

1. Why is the assessor's report allowed to conclude "uncertain" while the regulator's record may not?
2. What is the difference between a condition and a recommendation?
3. Why do real agencies publish assessment reports but keep deliberations separate?
4. Name one way stakeholder input legitimately changes an ERA.

Instructor answer key

- Q1: Typically refuge enforceability (it undercuts the resistance-management story); the public-interest reviewer's checklist should force it onto the record.
- Q2: e.g., buffer strips on developer-managed seed-production fields, or binding monitoring triggers with automatic suspension — conditions that do not rely on distributed compliance.
- Q3: Common honest answers: acceptable-risk threshold for the wild-relative pathway; whether economic effects on smallholders count as "consequence." Both are policy judgments; grading rewards naming them.
- Q4: Legitimately: local knowledge of wild relatives/land use can refine exposure characterization; values can set decision thresholds — but not edit the hazard table.

Advanced challenge

Rewrite the decision for a *different* jurisdiction (choose: US Coordinated Framework, EU Directive 2001/18, Pakistan NBC) with the same dossier. List exactly which parts of the assessment carried over unchanged and which decision elements changed — the empirical map of science vs policy in this course.

Previous: [Lab 10 — Uncertainty and Sensitivity Analysis](Lab-10-Uncertainty-Sensitivity-Analysis.md) ·

Next: [Lab 12 — Capstone Environmental Risk

Assessment](Lab-12-Capstone-Environmental-Risk-Assessment.md)

Lab 12 — Capstone: GMO Environmental Risk Assessment

Activity type: Capstone Project (student-led, multi-week)

Duration: 6–8 weeks (milestones) · **Level:** Advanced (course synthesis)

Navigation: [↑ Course Home](#) · [← Previous Lab](#) · [→ Assessment Package](#)

■ Builds on: every module; template below · ■ Data: DATA/ · ■ Rubric: Section 9

Learning objectives

On completing the capstone a student can independently:

1. Select a real GMO and define the scope of a conceptual ERA for a specified receiving environment.
2. Execute the full Module 3 workflow from problem formulation to risk communication.
3. Integrate course datasets (Labs 03–10) as quantitative backbones for their own assessment.
4. Produce a professional ERA report and defend it orally.

Background

This is the course's synthesis exercise: every module contributes one section. The chosen GMO must be **real** (a documented event/product); the receiving environment must be **specified** (not "the world"); all quantitative treatment uses the course's clearly labelled simulated data or published literature values with citations. The capstone is *conceptual* — no laboratory work, no field release, no real dossier data.

Scientific principle

The capstone must demonstrate the course's core competence: for every hazard claimed, the student shows the chain

Trait + Organism + Environment + Exposure + Evidence + Uncertainty → Risk statement

— and for every risk statement, at least one management/monitoring instrument (Module 18–19).

Materials/data

- Course modules, datasets, and labs (all reusable with citation).
- Authoritative literature for the chosen GMO (see REFERENCES.md); regulator summaries are the most efficient sources.
- Report template (Section 8), presentation requirements (Section 7), rubric (Section 9).

Safety

Conceptual exercise; no biosafety measures required. Intellectual-safety rule from Lab 11 applies: no invented data; uncertainty is carried, not hidden.

Step-by-step workflow

1. **Week 1 — Selection & scope.** Choose a real GMO + receiving environment; submit a one-page problem formulation for instructor sign-off (guardrail: feasibility).
2. **Week 2–3 — Hazard identification.** Apply Lab 01's worksheet; produce the hazard table with mechanisms and exposure pathways (Modules 4–13 logic).
3. **Week 3–4 — Characterization.** Reuse/adapt Lab 03 (gene flow), Lab 05 (resistance if applicable), Lab 07 (composition if food/feed relevant), Lab 08 (dose–response if a protein is involved), Lab 09 (matrix),

Lab 10 (uncertainty) — at least **four** quantitative elements must appear.

4. **Week 5 — Management & monitoring.** For each characterized risk: instrument, trigger, residual risk (Modules 18–20).

5. **Week 6 — Communication products.** Executive summary (≤ 1 page, lay-audience version); full report; presentation.

6. **Week 7–8 — Defence.** 15-minute presentation + 15-minute viva (questions drawn from the Assessment package style).

Data tables

The report must include (at minimum):

- Hazard-identification table (Lab 01 format).
- Risk-characterization table (Lab 11 format).
- Risk matrix figure (Lab 09).
- Uncertainty/sensitivity summary (Lab 10 format: rank table + one tornado figure).
- Monitoring-trigger table (Module 19 format).

Calculations

As required by the chosen GMO — minimum four quantitative elements from Step 3, each with stated assumptions and limitations.

Expected results

There is no single correct conclusion. Graded quality signals:

- Hazards specific to the *chosen* organism and *specified* environment (boilerplate hazards indicate failure).
- Uncertainty stated as intervals/flags, not adjectives.
- Management instruments matched to characterized risks (no orphan risks; no instrument-free "acceptables").
- The science/policy boundary respected throughout (Lab 11 discipline).

Interpretation

The final evidence-based conclusion must state: what is known, with what confidence; what is uncertain, and why it matters; what management closes the gap; and what monitoring would detect error. A conclusion of the form "safe" or "unsafe" without this apparatus does not pass.

Troubleshooting (instructor guidance)

Symptom	Intervention
GMO chosen too broad ("Bt crops")	Narrow to one event/product and one country/region
No quantitative backbone	Map the four required labs explicitly in supervision meetings
Values leaking into characterization	Apply the Lab 11 role-separation test to their draft
Report reads as advocacy (either direction)	Return to Module 42-style objectivity checklist; require comparator citations

Questions (student self-check before submission)

1. Could a reader reconstruct *which environment* you assessed from your report alone?

2. Does every hazard table row trace to at least one cited piece of evidence or an explicit data gap?
3. Is each risk statement accompanied by an instrument?
4. Is your conclusion falsifiable — what monitoring result would change it?

Viva questions (instructor bank)

1. Justify your receiving-environment choice — what in it drove your highest-ranked risk?
2. Which two parameters in your quantitative backbone dominate your conclusions, and what would better measurement change?
3. Where did you decide something that is a policy question, and how did you flag it?
4. Which of your conclusions is most sensitive to comparator choice?
5. If your top-ranked monitoring trigger fired in year two, what happens next?

Instructor answer key

No fixed answers; the rubric operationalises quality. For viva Q4, strong answers name a comparator study (e.g., against sprayed conventional baseline vs unsprayed) and articulate how the risk statement would shift. For Q5, strong answers connect the trigger to a pre-agreed adaptive-management action with a decision owner.

Advanced challenge

Add a "regulatory transferability" annex: run your assessment a second time under a different framework (Module 22) and document exactly which conclusions are framework-invariant. Defend the invariance claim — it is the deepest test of the course's central distinction.

Report template (Section 8 — required structure)

1. Executive summary (≤ 1 page, plain language)
2. GMO description & genetic modification (Modules 1, 4)
3. Intended trait & mechanism
4. Receiving environment specification (Module 2)
5. Problem formulation (Module 3)
6. Hazard identification (table; Modules 4–13)
7. Exposure pathways (Module 13; figure required)
8. Quantitative characterizations (≥ 4 elements; Labs 03/05/07/08/09/10)
9. Uncertainty analysis (Module 16; Lab 10 outputs)
10. Risk characterization (table + matrix; Module 17)
11. Risk management (instrument per risk; Module 18)
12. Monitoring plan (baseline, indicators, triggers; Module 19–20)
13. Risk communication summary (Module 21 — the lay executive summary)
14. Final evidence-based conclusion
15. References (AUTHORITATIVE SOURCES ONLY — no fabricated citations)

Assessment rubric (Section 9 — 100 points)

Criterion	Points	Indicators of full marks
Problem formulation & scope	10	Sharp, feasible, specified environment

Criterion	Points	Indicators of full marks
Hazard identification quality	15	Specific, mechanistic, traceable to evidence
Quantitative backbone (≥ 4 elements)	20	Correct method, stated assumptions, honest intervals
Uncertainty treatment	15	Separated variability/uncertainty; sensitivity-informed
Risk characterization & matrix	10	Evidence-anchored placements; no false precision
Management & monitoring integration	15	Every risk has an instrument; triggers pre-agreed
Communication products	10	Lay summary accurate without sensationalism or false reassurance
Objectivity & science/policy discipline	5	Lab 11 boundary maintained; no advocacy

Presentation requirements (Section 7): 15 minutes, 10–14 slides, must include the pathway figure, the risk matrix, and the monitoring-trigger table; the audience includes a "regulator" and a "reviewer" who each ask one question from the viva bank.

Previous: [Lab 11 — Case-Study Regulatory Assessment](Lab-11-Case-Study-Regulatory-Assessment.md)
 · Next: [Assessment Package](../ASSESSMENT/MCQs.md)

Practical Workbook — GMO Environment and Risk Assessments

Instructor: Dr. Saira Azam · Assistant Professor · Centre of Excellence in Molecular Biology · University of the Punjab, Lahore.

A compendium of all 12 practicals — the complete laboratory/practical program. This workbook is generated as PDF (PDF/Lab-workbook.pdf); each lab also exists as a standalone Markdown file in `LAB/`.

How to use this workbook

Activity type	Meaning
Wet Lab	Hands-on laboratory work with living material or chemicals
Demonstration	Instructor-performed; students observe and record
Simulation	Model-based exercise (software or worksheet)
Computational Exercise	Student analyses data in Python/spreadsheet
Case Study	Paper exercise on real documented scenarios

This course is predominantly non-wet-lab by design: environmental risk assessment is practised through data, models, and structured reasoning. No biosafety-critical materials are used anywhere in the program.

Practical program at a glance

#	Practical	Type	Duration	Course modules served
01	Terminology and Hazard Identification	Case Study / Simulation	2 h	2, 4–13
02	Risk-Assessment Workflow	Simulation	2 h	3
03	Gene-Flow Data Analysis	Computational	2.5 h	6, 24
04	Non-Target Data Analysis	Computational	2.5 h	8, 24
05	Resistance Evolution Simulation	Simulation (computational)	3 h	11, 25
06	Environmental Exposure Analysis	Computational	2.5 h	13, 24
07	Compositional Assessment	Computational / Case Study	2.5 h	5, 14, 24
08	Dose–Response Analysis	Computational	2.5 h	15, 24
09	Risk-Ranking Matrix	Computational / Case Study	2 h	17, 2
10	Uncertainty and Sensitivity Analysis	Computational	3 h	16, 17, 24
11	Case-Study Regulatory Assessment	Case Study (role-play)	3 h	22, 23
12	Capstone Environmental Risk Assessment	Capstone Project	6–8 weeks	all

Every practical includes: learning objectives · background · scientific principle · materials/data · safety · step-by-step workflow · data tables · calculations · expected results · interpretation · questions · viva

questions · instructor answer key · advanced challenge.

Datasets used

All quantitative practicals draw on the course's **clearly labelled simulated datasets** in `DATA/` (10 exercises, seeded and regenerable via `generate_datasets.py`; documented answers reproducible via `analysis_demo.py`). See the DATA README for file inventory.

Recommended scheduling

- **Weeks 1–2:** Labs 01–02 (foundations, framework)
- **Weeks 3–5:** Labs 03–08 (quantitative toolkit)
- **Weeks 6–7:** Labs 09–10 (characterization and uncertainty)
- **Week 8:** Lab 11 (regulatory role-play)
- **Weeks 9–16:** Lab 12 (capstone, alongside final modules)

General safety statement

No living GMOs, pathogenic material, chemical hazards, or fieldwork are involved in any practical. The professional-safety rules that *do* apply everywhere: (1) never present simulated data as real; (2) never state conclusions your evidence cannot carry; (3) cite comparators explicitly; (4) carry uncertainty honestly.

Instructor: Dr. Saira Azam — Centre of Excellence in Molecular Biology, University of the Punjab, Lahore.